

MH2500 Probability and Introduction to Statistics

Slides for Week 11-1

Topics: Moment Generating Functions

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Moment-Generating Functions

Moment-Generating Function is a very useful tool that can simplify certain calculations dramatically.

The **moment-generating function (mgf)** of a random variable X is

$$M(t) = \mathbb{E}(e^{tX}),$$

if the expectation is defined.

$$M(t) = \begin{cases} \sum e^{tx} p(x), & \text{discrete case} \\ \int_{\mathbb{R}} e^{tx} p(x) dx, & \text{continuous case.} \end{cases}$$

- $M(t) = \mathbb{E}(e^{tX})$ may or may not exist for a particular value of t .

Properties of MGF

Theorem

Suppose the moment generating function exists for t in a neighbourhood of 0 (an open interval containing 0). Then it *uniquely* determines the probability distribution.

Proof is beyond the scope of this course.

Differentiate $M(t)$ is

- X is discrete
- X is continuous

MGF of Binomial

Find the mgf of $X \sim \text{Bin}(n, p)$. Use it to find $\mathbb{E}(X)$ and $\text{Var}(X)$.

$$\begin{aligned} M(t) &= \sum_{k=0}^n e^{tk} \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=0}^n \binom{n}{k} (pe^t)^k (1-p)^{n-k} \\ &= (pe^t + 1 - p)^n \end{aligned}$$

The sum converges for all t . By differentiating, we have

$$M'(t) = n(pe^t + 1 - p)^{n-1} pe^t$$

$$M''(t) = n(n-1)(pe^t + 1 - p)^{n-2} (pe^t)^2 + n(pe^t + 1 - p)^{n-1} pe^t.$$

Evaluating these derivatives at $t = 0$ gives

$$\mathbb{E}(X) = np, \quad \mathbb{E}(X^2) = n(n-1)p^2 + np,$$

and so $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E} X)^2 = np(1-p)$.

MGF of Poisson

Find the mgf of $X \sim \text{Poisson}(\lambda)$. Use it to find $\mathbb{E}(X)$ and $\text{Var}(X)$.

$$\begin{aligned} M(t) &= \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} e^{-\lambda} \\ &= e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}. \end{aligned}$$

The sum converges for all t . Differentiating, we have

$$\begin{aligned} M'(t) &= \lambda e^t e^{\lambda(e^t - 1)} \\ M''(t) &= \lambda e^t e^{\lambda(e^t - 1)} + \lambda^2 e^{2t} e^{\lambda(e^t - 1)}. \end{aligned}$$

Evaluating these derivatives at $t = 0$ gives

$$\mathbb{E}(X) = \lambda, \quad \mathbb{E}(X^2) = \lambda^2 + \lambda,$$

and so $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E} X)^2 = \lambda$.

When X is continuous

and

$$M'(t) = \frac{d}{dt} \int_{-\infty}^{\infty} e^{tx} p(x) dx = \int_{-\infty}^{\infty} x e^{tx} p(x) dx$$

$$M''(t) = \frac{d}{dt} \int_{-\infty}^{\infty} x e^{tx} p(x) dx = \int_{-\infty}^{\infty} \boxed{x p(x) \cdot x} e^{tx} dx$$

$$M'(0) = \int_{-\infty}^{\infty} x p(x) dx = \mathbb{E}(X).$$

$$M''(0) = \int_{-\infty}^{\infty} x p(x) \cdot x dx = \int_{-\infty}^{\infty} x^2 p(x) dx = \mathbb{E}(X^2)$$

In general,

By differentiating $M(t)$ r times, we have

$$M^{(r)}(0) = \mathbb{E}(X^r).$$

Theorem

If the moment generating function exists in a neighbourhood of 0, then so do all the moments and $M^{(r)}(0) = \mathbb{E}(X^r)$.

MGF of Exponential

Find the mgf of $X \sim \text{Exp}(\lambda)$. Use it to find $\mathbb{E}(X)$ and $\text{Var}(X)$.

$$M(t) = \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx = \frac{\lambda}{\lambda - t}, \quad t < \lambda.$$

By differentiating, we have

$$M'(t) = \frac{\lambda}{(\lambda - t)^2}, \quad M''(t) = \frac{2\lambda}{(\lambda - t)^3}.$$

Evaluating these derivatives at $t = 0$ gives

$$\mathbb{E}(X) = \frac{1}{\lambda}, \quad \mathbb{E}(X^2) = \frac{2}{\lambda^2}, \quad \text{Var}(X) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

MGF of Standard Normal

Find the mgf of $X \sim N(0, 1)$. Use it to find $\mathbb{E}(X)$ and $\text{Var}(X)$.

$$M(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-1/2x^2+tx} dx.$$

The integral converges for all t and can be evaluated using the technique of completing the square.

$$-\frac{x^2}{2} + tx = -\frac{1}{2}(x - t)^2 + \frac{t^2}{2}.$$

Therefore,

$$M(t) = \frac{e^{t^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2} dx = e^{t^2/2} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2} dx \right) = e^{t^2/2}.$$

We just obtained

$$M(t) = e^{t^2/2}.$$

By differentiating, we have

$$M'(t) = te^{t^2/2}, \quad M''(t) = e^{t^2/2} + t^2 e^{t^2/2},$$

and

$$\mathbb{E}(X) = M'(0) = 0, \quad \mathbb{E}(X^2) = M''(0) = 1.$$

Hence

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = 1.$$

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Slides for Week 12-1

Topics: MGF continued

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Theorem

If X has the mgf $M_X(t)$ and $Y = a + bX$, then Y has mgf

$$M_Y(t) = e^{at} M_X(bt).$$

Proof:

$$\begin{aligned} M_Y(t) &= \mathbb{E}(e^{tY}) \\ &= \mathbb{E}(e^{t(a+bX)}) \\ &= \mathbb{E}(e^{at+btX}) \\ &= \mathbb{E}(e^{at} e^{btX}) \\ &= e^{at} \mathbb{E}(e^{btX}) \\ &= e^{at} M_X(bt). \end{aligned}$$

MGF of General Normal Distribution

From the earlier example, for $Z \sim N(0, 1)$, $M_Z(t) = e^{t^2/2}$.

Let $X \sim N(\mu, \sigma^2)$. Then $X = \mu + \sigma Z$ and

$$M_X(t) = e^{\mu t} M_Z(\sigma t) = e^{\mu t} e^{\sigma^2 t^2/2} = e^{\mu t + \sigma^2 t^2/2}.$$

By differentiating, we have

$$M'_X(t) = (\mu + \sigma^2 t) e^{\mu t + \sigma^2 t^2/2},$$

$$M''_X(t) = (\sigma^2) e^{\mu t + \sigma^2 t^2/2} + (\mu + \sigma^2 t)^2 e^{\mu t + \sigma^2 t^2/2}.$$

This gives

$$\mathbb{E}(X) = M'_X(0) = \mu, \quad \mathbb{E}(X^2) = M''_X(0) = \sigma^2 + \mu^2,$$

and hence $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sigma^2$.

Theorem

If X and Y are independent random variables with mgf's M_X and M_Y and $Z = X + Y$, then $M_Z(t) = M_X(t)M_Y(t)$ on the common interval where both mgf's exist.

Proof:

$$M_Z(t) = \mathbb{E}(e^{tZ}) = \mathbb{E}(e^{t(X+Y)}) = \mathbb{E}(e^{tX} e^{tY})$$

by the assumption of independence,

$$= \mathbb{E}(e^{tX}) \mathbb{E}(e^{tY}) = M_X(t) M_Y(t).$$

Sum of independent Poisson variables

Suppose X is Poisson with parameter λ and Y is Poisson with parameter μ , and that X and Y are independent random variables. Then $X + Y$ is Poisson with parameter $\mu + \lambda$.

We know that X and Y have mgf

$$M_X(t) = e^{\lambda(e^t - 1)} \quad M_Y(t) = e^{\mu(e^t - 1)}.$$

Hence

$$M_{X+Y}(t) = M_X(t)M_Y(t) = e^{(\lambda+\mu)(e^t - 1)}.$$

Therefore, $X + Y$ is Poisson with parameter $\lambda + \mu$.

Sum of independent Normal variables

Suppose $X \sim N(\mu, \sigma^2)$ and $Y \sim N(\nu, \tau^2)$ and are independent of each other.

- Find the mgf of $X + Y$.

Solution:

$$\begin{aligned} M_{X+Y}(t) &= M_X(t)M_Y(t) = \left(e^{\mu t + \sigma^2 t^2 / 2} \right) \left(e^{\nu t + \tau^2 t^2 / 2} \right) \\ &= e^{(\mu + \nu)t + (\sigma^2 + \tau^2)t^2 / 2}. \end{aligned}$$

Thus $X + Y \sim N(\mu + \nu, \sigma^2 + \tau^2)$.

Remarks

- 1 If X and Y have a joint distribution, their **joint mgf** is defined as

$$M_{XY}(s, t) = \mathbb{E}(e^{sX+tY}).$$

- 2 It can be shown that **two random variables X and Y are independent if and only if their joint mgf factors into the product of the mgf's of the marginal distributions.**
- 3 A major limitation of the mgf is that it may not exist.